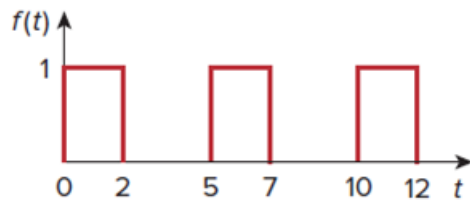


Problem

Determine the Laplace transform of the periodic function in Fig. 15.8.

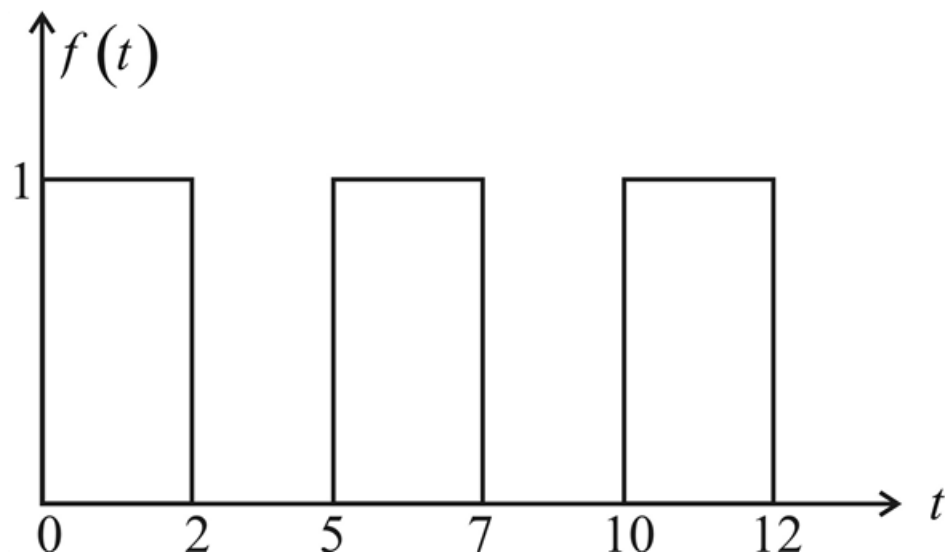
Figure 15.8:



Step-by-step solution

Step 1 of 2

Given periodic function is



Step 2 of 2

The period of the function is $T = 5$

The transform of the first period of the function

$$f_1(t) = u(t) - u(t-2)$$

Using the time –shift property

$$F_1(s) = \frac{1}{s} - \frac{e^{-2s}}{s}$$

$$F_1(s) = \frac{1 - e^{-2s}}{s}$$

Thus the transform of the periodic function

$$F(s) = \frac{F_1(s)}{1 - e^{-Ts}}$$

$$\therefore F(s) = \frac{1 - e^{-2s}}{s(1 - e^{-5s})}$$